Homework 05
REAL VARIABLES I
Fall 2026

Due: 2026-09-27 19:00
Topics: Borel sets and measurable functions

Instructions
Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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1. (Countable suprema and infima)

IN CLASS

Let $(X, \mathcal{A})$ be a measurable space and let $(f_n : X \to \bar{\mathbb{R}})_{n\in\mathbb{N}}$ be measurable.

1.a (weight: 0.2) Prove directly from a level-set criterion that the functions

$$x \longmapsto \sup_{n\in\mathbb{N}} f_n(x) \qquad \text{and} \qquad x \longmapsto \inf_{n\in\mathbb{N}} f_n(x)$$

are measurable. Be explicit about where countability is used.

1.b (weight: 0.2) Let $\mathcal{J}$ be an arbitrary index set and let $(f_i : \mathbb{R} \to \mathbb{R})_{i\in\mathcal{J}}$ be Borel measurable. Provide an example where the pointwise supremum

$$x \longmapsto \sup_{v\in V} f_v(x)$$

is not Borel measurable.

1.c (weight: 0.3) There is a useful positive result despite the preceding example. Let I be an arbitrary nonempty index set and let $(g_i : \mathbb{R} \to \mathbb{R})_{i\in I}$ be continuous. Prove that the extended-real-valued function

$$g(x) \coloneqq \sup_{i\in I} g_i(x)$$

is Borel measurable, even when I is uncountable.

1.d (weight: 0.3) IN CLASS
Let $g : (X, \mathcal{E}) \to [0, +\infty]$ be a Borel measurable function. Prove that it is the pointwise monotone limit of a sequence g_n of simple functions.

Hint. Fix n and discuss when $g(x) \in [k2^{-n}, (k+1)2^n]$

2. (Graphs of continuous and Borel functions)

For a function $f\colon \mathbb{R} \to \mathbb{R}$, define

$$\operatorname{graph}(f) \coloneqq \{(x, y) \in \mathbb{R}^2 : y = f(x)\}.$$

2.a (weight: 0.4) Suppose that f is continuous. Prove that $\operatorname{graph}(f)$ is a Borel subset of $\mathbb{R}^2$.

2.b (weight: 0.6) Prove the stronger statement that $\operatorname{graph}(f)$ is Borel in $\mathbb{R}^2$ whenever $f\colon \mathbb{R} \to \mathbb{R}$ is Borel measurable.

3. (First crossing and infinitely many crossings)

Let $(f_n : X \to \mathbb{R})_{n\in\mathbb{N}}$ be measurable and fix $a \in \mathbb{R}$. Define $\tau\colon X \to \mathbb{N} \cup \{\infty\}$ by

$$\tau(x) \coloneqq \begin{cases} \min\{n \in \mathbb{N} : f_n(x) > a\}, & \text{if this set is nonempty,} \\ \infty, & \text{otherwise.} \end{cases}$$

Give $\mathbb{N} \cup \{\infty\}$ its full σ-algebra.

3.a (weight: 0.6) Prove that τ is measurable.

3.b (weight: 0.4) Prove that the set

$$\{x \in X : f_n(x) > a \text{ for infinitely many } n\}$$

is measurable.